

**Reforming Pay-As-You-Go Pension
Systems
in Aging Economies:
Italy, Japan, Greece, Germany, and France**

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Abstract

This paper examines the policy levers used by governments to reform public pay-as-you-go (PAYG) pension systems in response to demographic aging. Drawing on the pension reform histories of Italy, Japan, Greece, Germany, and France—five countries sharing pronounced demographic pressures and a predominant first-pillar public system—the paper documents the main restrictive interventions implemented since the 1990s and classifies them into a common taxonomy. The analysis finds that all five countries have relied on parametric reforms, including increases in statutory and contributory retirement ages, reductions in benefit generosity through modified indexation and benefit formulas, and increases in contribution rates or general-revenue transfers. Several countries also undertook structural reforms, shifting from defined-benefit to notional defined-contribution architectures. The paper highlights the growing role of automatic adjustment mechanisms as a tool for insulating pension finances from demographic and macroeconomic shocks.

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Introduction

The global population reached eight billion in 2022 and is projected to grow to approximately 9.7 billion by 2050. This growth, however, is driven almost entirely by developing nations, while Europe, North America, South Korea, and Japan have already reached demographic stagnation or are in outright decline. Life expectancy at birth has risen markedly, reaching 72.8 years in 2019—nearly nine years higher than in 1990. The share of the population aged 65 and above is set to increase from 10% in 2022 to 16% by 2050, placing mounting pressure on pay-as-you-go (PAYG) pension systems (United Nations, Department of Economic and Social Affairs, Population Division 2024). These systems, which finance current pension benefits with contributions from the active workforce, face fundamental challenges from falling fertility and rising longevity (European Commission 2023).

Some countries are affected more acutely than others. Italy, Japan, South Korea, and much of Europe face a scenario in which the ratio of retirees to workers could double in the coming decades. Many of these countries—characterized by a predominant public PAYG pillar—have intervened to reform otherwise unsustainable systems since the 1990s (OECD 2023), broadly consistent with the early prescriptions of World Bank (1994).

Research question. This paper asks: what policy levers have governments used to reform the public first pillar of pension provision in aging economies, and how can these interventions be systematically classified? The analysis covers Italy, Japan, Greece, Germany, and France—five countries selected on the basis of their demographic comparability and the predominance of public PAYG systems in their retirement income architecture.

Contribution and structure. The paper contributes a cross-country survey of restrictive pension reform measures implemented from the late twentieth century onward, drawing on primary legislation and a diverse body of comparative pension scholarship. Chapter 1 introduces the mechanics of PAYG systems and documents the demographic context across the five countries, using data from the United Nations (United Nations, Department of Economic and Social Affairs, Population Division 2024) and the OECD (OECD 2023). Chapters 2 through 6 examine each country in turn, tracing the historical origins of its pension system before describing the main restrictive reforms chronologically. Chapter 7 synthesizes the evidence into a taxonomy of reform instruments. Chapter 8 concludes.

1 The Pay-As-You-Go System and the Demographic Context

1.1 The Pay-As-You-Go System

In pay-as-you-go (PAYG) pension systems, current contributions are used to finance current pension benefits. Unlike funded systems—often referred to as *fully-funded* or capitalization systems—PAYG arrangements do not systematically accumulate reserves in individual accounts. The single-period budget constraint that governs a PAYG system is given by Equation (1), which requires that total contributions equal total pension expenditure:

$$N_p \times P = \alpha \times N_L \times R \implies \alpha = \frac{P}{R} \times \frac{N_p}{N_L} \quad (1)$$

where N_p and N_L denote the number of retirees and workers respectively, P is the average pension benefit, R is the average gross wage, and α is the equilibrium contribution rate. This rate depends on two key ratios: the old-age dependency ratio (N_p/N_L) and the replacement rate (P/R). In the literature, α serves as a standard proxy for the financial sustainability of a PAYG system: a rising equilibrium contribution rate signals either population aging—reflected in a worsening dependency ratio—or an imbalance between wage and pension levels.¹ As shown in Cipriani (2014) and Fanti and Gori (2012), demographic aging is a central determinant of this equilibrium, a point formalized in the decomposition proposed by the European Commission (European Commission 2023, p. 48) and reported in Equation (2).

$$\begin{aligned} \frac{PE}{Y} = & \underbrace{\frac{N_{65+}}{N_{20-64}}}_{\text{dependency ratio}} \times \underbrace{\frac{N_p}{N_{65+}}}_{\text{coverage ratio}} \\ & \times \underbrace{\frac{P}{Y/H}}_{\text{benefit ratio}} \times \underbrace{\frac{N_{20-64}}{H}}_{\text{labour market effect}} \end{aligned} \quad (2)$$

where PE denotes total pension expenditure, Y is gross domestic product (GDP), N_{65+} and N_{20-64} are the populations aged 65+ and 20–64 respectively, N_p is the number of pensioners, P is the average pension benefit, and H denotes total hours worked (by individuals aged 20–74). The decomposition highlights four distinct channels through which pension expenditure as a share of GDP evolves. The *dependency ratio* captures demographic pressure, i.e. the relative size of the elderly population. The *coverage ratio* measures the share of elderly individuals receiving a pension. The *benefit ratio* reflects the generosity of pensions relative to average labour productivity (proxied by GDP per hour worked). Finally, the *labour market*

¹A system can be maintained with an effective contribution rate above the equilibrium rate, accumulating reserves, or below it, requiring transfers from general taxation—a *pension deficit*.

effect captures labour utilisation, linking the working-age population to total hours worked in the economy.

1.2 Demography and Population Aging

The sustainability of PAYG systems is thus directly affected by population aging (OECD 2023). This process is common to all advanced economies, but bears most heavily on Europe, North America, Japan, and South Korea. Its two principal drivers are the long-run decline in fertility and the steady increase in longevity (United Nations, Department of Economic and Social Affairs, Population Division 2024).

Average fertility across OECD countries currently stands at 1.59 children per woman—well below the replacement threshold of 2.1—and is projected to stabilize at around 1.60 in the long run.² Life expectancy continues to rise for both men and women, with Japan, South Korea, France, Spain, and Italy recording some of the highest values globally, while old-age dependency ratios are highest in Japan, Italy, Greece, and Spain.

Among the five countries examined in this paper, Italy and Japan exhibit the lowest fertility rates as of 2022; Germany (1.53) and France (1.79) record somewhat higher values, though still below the replacement threshold; Greece occupies an intermediate position (1.38). As illustrated in Figure 1, fertility has declined across all five countries between 1962 and 2022. With the exception of France—which is projected to remain around 1.75 children per woman—a modest recovery is expected over the next several decades, converging toward approximately 1.5 children per woman.

²The replacement threshold of 2.1 is defined by the OECD as the fertility rate at which the population remains constant, assuming zero net migration and no change in mortality (OECD 2023).

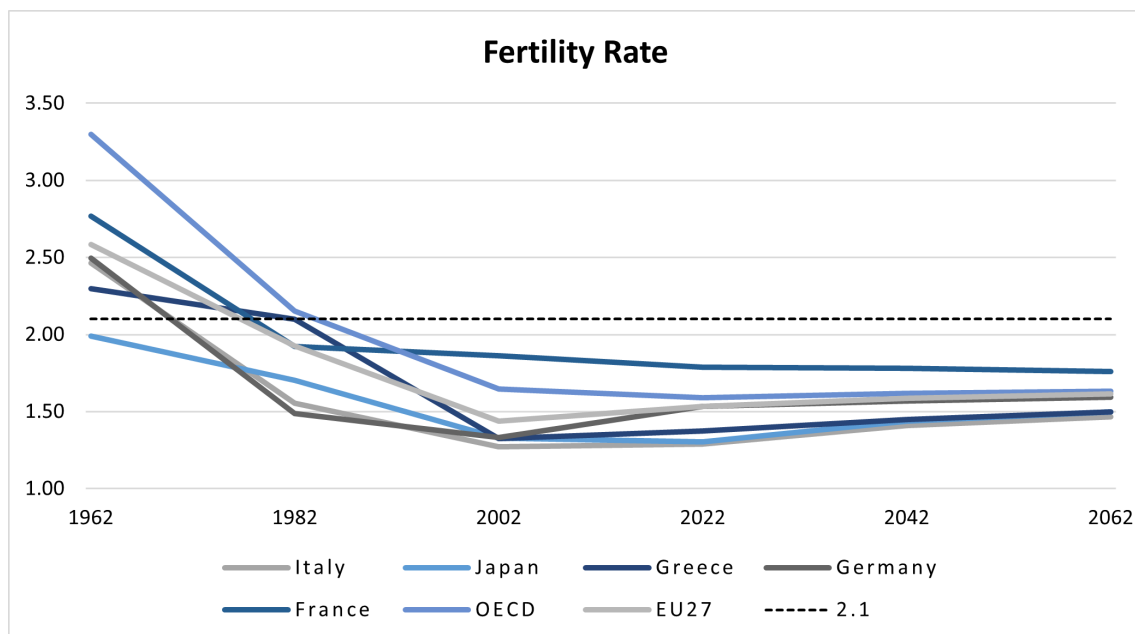


Figure 1. Total fertility rates for Italy, Japan, Greece, Germany, France, the OECD average, and the EU average. Historical series through 2023, median scenario thereafter. Source: OECD (2023).

Alongside subdued fertility, aging in these countries is compounded by life expectancy levels that consistently exceed the OECD and EU averages. Japan, Italy, and France are among the leaders in both male and female life expectancy (Figure 2), and the upward trend is projected to continue, with a gain of roughly nine years by 2100 relative to 2024 (Figure 3).

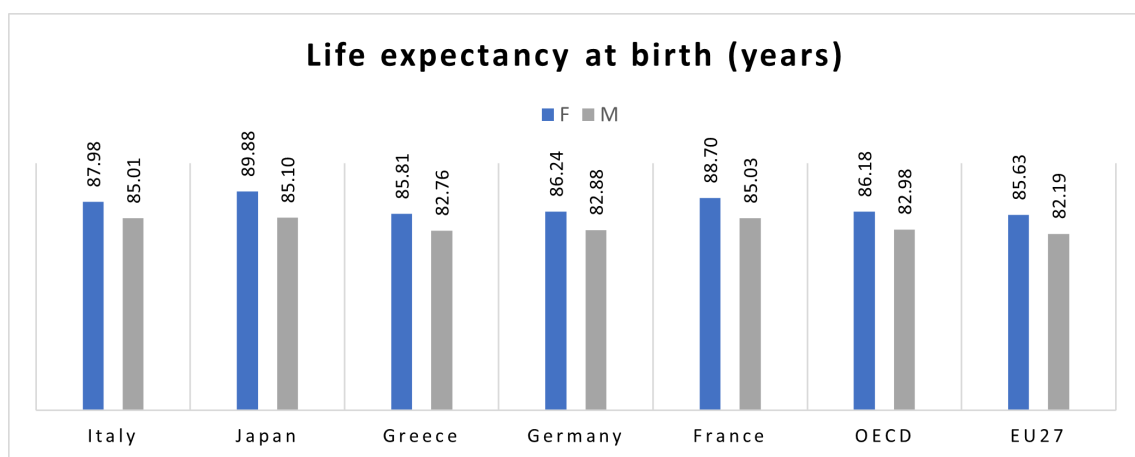


Figure 2. Life expectancy at birth. Source: OECD (2023).

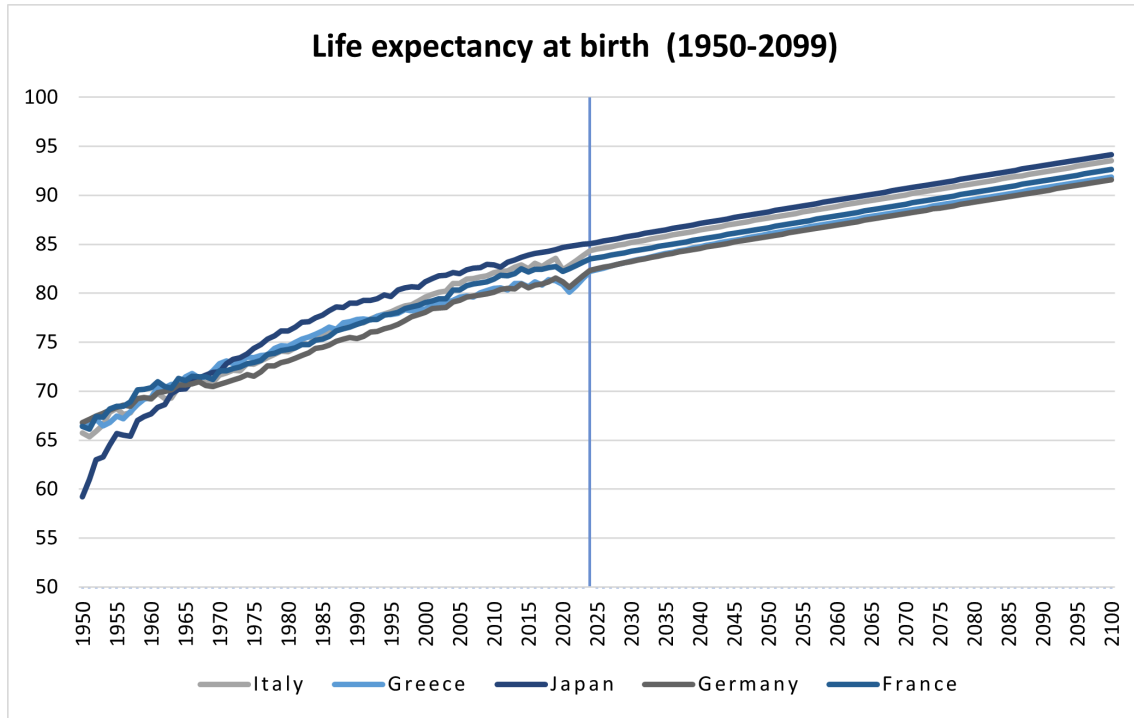


Figure 3. Life expectancy at birth. Source: OECD (2023).

The aging dynamic is further confirmed by the old-age dependency ratio.³ Between 1952 and 2022 this ratio at least doubled in all five countries. In Greece and Italy it tripled; in Japan it increased nearly sixfold, from 9.95% to 55.43%. As of 2022, excluding Japan, the remaining four countries cluster between 38% and 41%.

Looking ahead, the old-age dependency ratio is projected to double again by 2082, reaching approximately 86% in Japan, 83% in Italy, and 80% in Greece, with France and Germany expected to record values between 60% and 65%. Expressed as workers per retiree, projections imply 1.16 workers per retiree in Japan by 2082, 1.2 in Italy and Greece, and roughly 1.5 in Germany and France—compared with current values ranging from 1.8 (Japan) to 2.6 (Germany).

³Defined here as the ratio of the population aged 65 and over to the population aged 20–64, following OECD (2023).

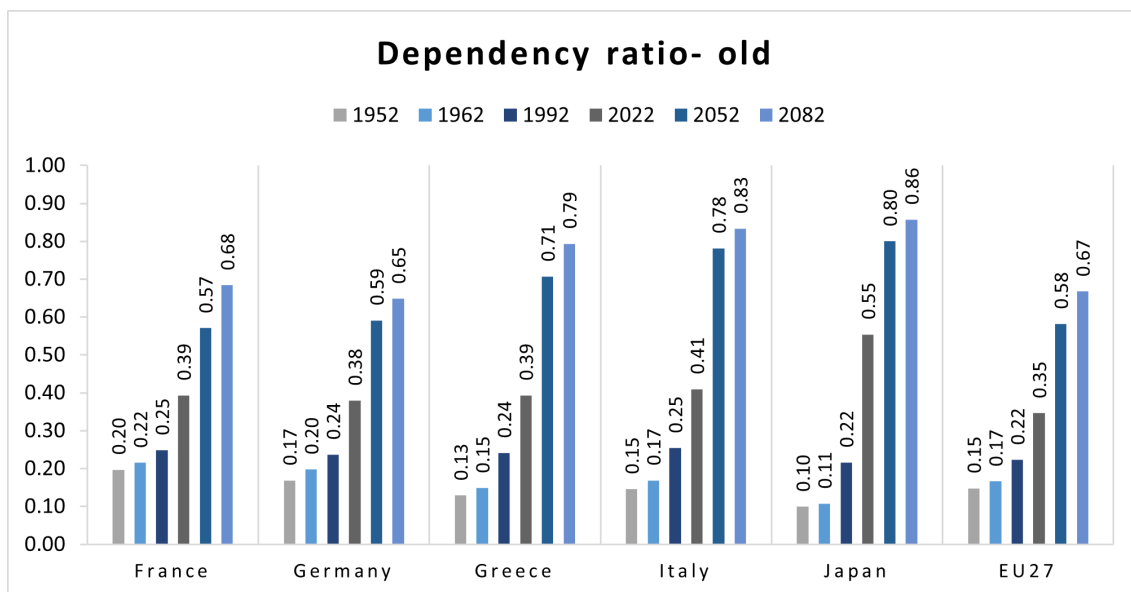


Figure 4. Trend of dependency ratio - old (65+ vs working age) Source: OECD (2023).

The total population of these countries has either already peaked or is approaching its maximum. Italy, Greece, and Japan belong to the first group, having reached their peaks in 2014, 2007, and 2011 respectively; all three are projected to lose roughly 40% of their 2022 population by 2100. Germany is near its own peak and is projected to shrink by around 17%. France presents a different trajectory: it is expected to reach its population maximum only in the 2040s and subsequently decline at a much slower rate of approximately -0.06% per year.

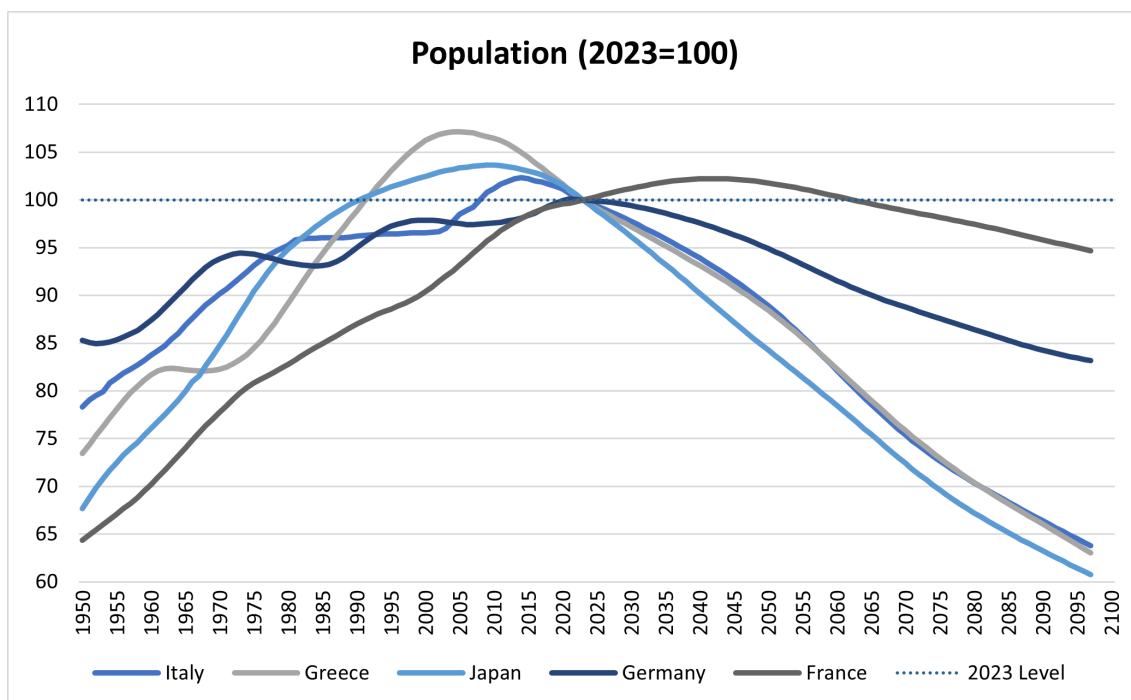


Figure 5. Population projections, 2023 = 100 Source: OECD (2023).

Summary. These data support a two-group classification of the countries under analysis. The first group—Japan, Italy, and Greece—already experiences pronounced demographic change: low and declining fertility, above-average longevity, high old-age dependency ratios, and absolute population contraction. The second group—Germany and France—faces aging pressures that are either less intense or later in onset. Germany’s dependency ratio has grown more gradually, while France’s persistently higher fertility rate (above 1.75 throughout the projection period) places it in a comparatively more favorable position.

2 Italy

Italy's pension system has undergone three structural transformations since 1992, moving from a generous defined-benefit architecture toward a notional defined-contribution framework, while also navigating an expansionary reversal in the years following the Fornero reform. The central thread running through these reforms is the interplay between demographic pressures, fiscal constraints, and external economic commitments.

2.1 Origins of the Italian Pension System

The roots of the Italian public pension system trace back to the early years of the unified Italian state.⁴ The first steps toward a social insurance framework were taken in 1861, and by the turn of the twentieth century the first mutual fund schemes for workers had been established. Over the following six decades, participation in old-age protection programs became mandatory for all workers, eventually covering some twelve million insured persons. During the fascist period, the modern national pension institute—known today as INPS—was created under the name INFPS. In the postwar decades, especially during the 1950s and 1960s, both the coverage and the generosity of the system were progressively expanded. In 1965, an early-retirement pension based solely on years of contributions (35 years) was introduced; in 1969 the earnings-related benefit formula was adopted. The early 1970s saw the introduction of a seniority pension with especially permissive terms for public employees—the so-called *baby pension*—allowing retirement after as few as 14 years, 6 months, and one day of service for certain categories of female public workers.⁵

By the early 1990s the system was a pay-as-you-go defined-benefit (DB) scheme organized into three separate programs differentiated by occupational category.⁶ Benefits were calculated according to:

$$P = \beta \times R \times N \tag{3}$$

where $\beta = 0.02$, R was the pensionable wage, and N the number of contribution years at retirement. The definition of R differed across occupational categories: at the time of the Amato reform, it corresponded to the final salary for public employees,

⁴This section draws on Castellino (1995), Ferrera, Fargion, and Jessoula (2012), Jessoula (2009), and Jessoula (2019).

⁵Under Article 42 of Presidential Decree 1092, female public employees with children could retire after 14 years, 6 months, and one day; other public employees after 20 years; and local government employees after 25 years. The first disbursements occurred as early as 1981.

⁶Private-sector employees were enrolled in the *Fondo Pensioni Lavoratori Dipendenti* (FPLD); public employees had their own occupational schemes; the self-employed were covered by separate funds.

the average of the last five years for FPLD members, and the average of the last ten years for the self-employed.⁷ Retirement conditions also varied by sector and sex, as summarized in Table 1.

Table 1. Retirement conditions in Italy in the early 1990s. Source: adapted from Brugiavini and Fornero (1998, Table 2).

	Private employees		Public employees		Self-employed	
	M	F	M	F	M	F
Old-age retirement age	60	55	65	65	65	60
Contribution years (seniority)	35	35	20	15	35	35

2.2 The Amato Reform (1992)

In 1992, Italy ratified the Maastricht Treaty parameters—a 3% ceiling on the annual deficit-to-GDP ratio and a 60% ceiling on the debt-to-GDP ratio—creating immediate fiscal pressures. In the same year, speculative attacks on the lira forced a currency devaluation. This combination of European fiscal commitments and market pressure, described in the literature as the *vincolo esterno* (external constraint), has been a recurrent feature of Italian pension reform.⁸

Brugiavini and Fornero (1998) identify three structural weaknesses that motivated the intervention: (a) excessive generosity in benefit computation; (b) redistributive distortions across occupational categories; and (c) financial imbalance between contributions and aggregate pension expenditure. The Amato reform⁹ pursued two objectives (Castellino 1995): restoring financial balance and harmonizing the disparate occupational regimes.

The reform moved the public-sector employee scheme closer to the rules governing private employees and the self-employed. The minimum contribution requirement for a seniority pension was standardized at 35 years across all categories, eliminating the more favorable treatment previously available to public employees. The minimum contributory period for an old-age pension was raised from 15 to 20 years, with limited transitional exceptions.¹⁰ The benefit formula was also revised: the pensionable wage base was extended to the wage average over the entire career, revalued at the rate of inflation plus one percentage point—a significant change relative to the pre-reform reference periods.¹¹ Finally, pension indexation, previously linked to contractual

⁷Source: Brugiavini and Fornero (1998), Table 2.

⁸See, for example, Jessoula (2013) and Natali and Stamati (2015).

⁹Law No. 421/1992 and Legislative Decrees No. 503/1992 and No. 124/1993.

¹⁰Workers who had already accumulated 15 years of contributions by 1992 could retire under the pre-reform rules.

¹¹Each annual wage was capitalized at inflation plus one percentage point, excluding wages below 80% of the average (Castellino 1995).

minimum wages, was restricted to price inflation alone.

Despite these measures, Castellino (1995) notes that the reform left unresolved issues—most notably the persistently low effective retirement age—and that a further intervention would be required to address the long-run unsustainability of the system, the regressivity of the benefit formula, and the implicit incentives for early retirement (Jessoula and Raitano 2015).

2.3 The Dini Reform (1995)

The Dini reform represented a structural break with the previous system. Its stated objectives were to stabilize pension expenditure relative to GDP, harmonize the rules across different funds, expand coverage, and facilitate the development of supplementary private provision (Jessoula and Raitano 2015).

The central innovation was the replacement of the defined-benefit formula with a notional defined-contribution (NDC) method—modeled on the Swedish system introduced in the same period (Palmer and Börsch-Supan 2006). Under the NDC method, a notional account accumulates the worker’s contributions—computed on the basis of pensionable wages multiplied by the relevant contribution rate—and is revalued annually at the rate of GDP growth.¹² At retirement, the notional stock is converted into an annuity using a transformation coefficient that depends on the worker’s age at retirement and life expectancy, independently of sex. This design creates strong actuarial incentives against early exit: a longer deferral raises the transformation coefficient and hence the annual benefit. Conversely, longer life expectancy reduces the coefficient, offsetting the cost of a longer payout period without increasing aggregate expenditure—a property known as “actuarial neutrality” (Jessoula and Raitano 2015).¹³

The seniority pension was phased out gradually, with different transitional arrangements by sector.¹⁴ The NDC rules were introduced in three tiers based on contributory seniority accrued by 1995: workers with at least 18 years of contributions were exempt from the new system; those with fewer than 18 years were subject to a pro-rata regime; and labor market entrants from 1996 onward fell entirely under the NDC. Full implementation was estimated to be complete only around 2050 (Brugiavini and Fornero 1998).

¹²The term “notional” reflects the fact that contributions credited to the account are not actual funds set aside in individual accounts, but hypothetical amounts consistent with PAYG financing. See Palmer and Börsch-Supan (2006).

¹³For each year of delayed retirement, the annual pension would increase by approximately 5–7% (Patriarca and Patriarca 2015).

¹⁴For private employees, two alternative paths were prescribed: a seniority pension after 35 years of contributions required a rising minimum age, while a pure seniority path required 40 years of contributions by 2008. The self-employed lost the seniority pension immediately, with a minimum age requirement of 57 from 1998. No gradual phaseout was established for the public sector.

2.4 Parametric Reforms: Maroni (2004) and Sacconi (2009–2010)

In 2004, further adjustments to retirement conditions were introduced through the Maroni reform. Starting in 2008, access to the seniority pension (35 contribution years) required a minimum age of 60, up from 57.¹⁵ The reform also reintroduced a gender distinction for seniority pensions (65 for men, 60 for women), which the European Commission subsequently challenged under Directive 96/97/EC; Italy was condemned by the Court of Justice in 2008 for maintaining discriminatory provisions under Article 141 of the Treaty (Jessoula 2010).¹⁶

The removal of this discriminatory provision was included in the first Sacconi package (2009), which introduced a gradual increase in the female statutory retirement age to 65, and implemented an automatic adjustment mechanism linking access conditions to life expectancy, effective from 2015. A second package (Sacconi II, 2010) brought forward the increase in women’s retirement age to 2012 in response to deteriorating economic conditions. These measures nonetheless failed to reassure financial markets, and the resulting loss of confidence contributed to the fall of the Berlusconi government and the appointment of the Monti technocratic government, which subsequently enacted the Fornero reform.

2.5 The Fornero Reform (2011)

Rising stress on Italian sovereign debt and mounting international pressure¹⁷ led to the replacement of the Berlusconi government with the Monti cabinet in late 2011. The government’s emergency fiscal package—*Salva Italia*—included a sweeping pension reform designed by the Minister of Labor, Elsa Fornero. The overarching objective was fiscal stabilization through a sharp increase in effective retirement ages and the formal abolition of the seniority pension, an institution considered anomalous in the Italian context (Fornero 2015).

The reform’s modifications can be organized into two categories: changes to retirement conditions (Table 2) and changes to the benefit formula (Table 3).

¹⁵The abrupt transition prompted the colloquial term *scalone* (“big step”) in public discourse.

¹⁶See Court of Justice, Case C-46/07: “The Court found that Italy, by maintaining in force legislation under which public employees have the right to a retirement pension at different ages depending on whether they are men or women, has failed to fulfil its obligations under Article 141 CE.”

¹⁷A notable example is the Draghi–Trichet letter to the Italian government, which explicitly called for further pension intervention, noting that it was possible to “intervene further in the pension system, [...] obtaining savings as early as 2012” (*Il Sole 24 Ore*, 2011).

Table 2. Statutory retirement ages under the Fornero reform.

	Pre-reform (2011)	2012	2013	2018 (projected)	2021 (min.)	At regime (≈2050)
Men (all sectors)	65	66	66y 3m	66y 7m	67	69y 9m
Women (public)	65 (61)	66	66y 3m	66y 7m	67	69y 9m
Women (private)	60	62	62	66y 7m	67	69y 9m

Note: The 2018 projected age was contingent on life expectancy developments.

The 2021 minimum of 67 was a floor, independent of life expectancy.

Table 3. Additional retirement conditions under the Fornero reform.

Workers subject only to the NDC system			
Regime	Age relative to statutory age	Age in 2011	Multiple of social allowance
Early pension	3 years early (age 63 in 2011)	63	2.8× (€1,170/month)
Deferred pension	4 years later (age 70 in 2011)	70	1.5× (€627/month)
Workers subject to the mixed (pro-rata) system			
Men	42 contribution years and 1 month		
Women	41 contribution years and 1 month		

Note: Contributory seniority for pro-rata workers is linked to life expectancy in the same manner as statutory retirement ages (cf. Table 2).

The seniority pension was abolished outright and replaced by the “early pension” (*pensione anticipata*). On the benefit side, the Fornero reform accelerated the pro-rata application of the NDC formula to workers who had previously been exempted. Additionally, pension indexation was suspended for pensions above €1,400 gross per month for 2012.¹⁸

2.5.1 A Contested Reform

Beyond the political controversy it generated,¹⁹ the Fornero reform was sharply criticized for the *esodati* problem. The sudden increase in retirement ages displaced

¹⁸This measure was subsequently declared unconstitutional by the Constitutional Court (Ruling No. 70/2015) (Fornero 2015).

¹⁹The political debate is not examined here; for a discussion, see Fornero (2015).

the expectations of workers who had already left the labor market in anticipation of imminent retirement. Two distinct groups were affected: those with at least 20 years of contributions²⁰ who met the financial threshold but not the new age requirements, and those who had not yet reached 20 years and were now required to accumulate additional contributions despite being disconnected from the labor market.

Strictly defined, *esodati* are workers who had left employment in anticipation of retirement but found themselves unable to retire under the new rules—either because they had insufficient savings to bridge the gap or because they lacked the contributory seniority to re-enter the benefit system (Nocella 2013). The reform included provisions to safeguard an estimated 65,000 such individuals, but this figure proved a serious underestimate: as of 2021, the affected population was estimated at over 200,000 (Camera dei Deputati – Servizio Studi 2021). A series of successive “safeguard” measures were enacted to address the situation.²¹

A further criticism concerns the compatibility of a mandatory minimum retirement age with the NDC architecture. In a pure NDC system, benefit levels adjust automatically through the transformation coefficient, making a statutory retirement floor redundant and potentially distortionary (Raitano 2013). Workers fully subject to the contributory system—particularly those with discontinuous careers, who are characteristic of younger cohorts—could face required working lives extending to age 74, with correspondingly low replacement rates (Jessoula 2014). This cohort will also contend with slower GDP growth, weaker labor markets, and less favorable macroeconomic conditions than previous generations, a phenomenon discussed in the literature as “Bad Start, Bad Finish” (Boeri and Galasso 2012; Takayama and Shiraishi 2012).

2.6 An Expansionary Phase (2016–2023)

Following the Fornero reform, a succession of measures sought to soften its conditions and ease access to retirement (Jessoula 2017). The Renzi–Poletti reform (2016) introduced measures of an expansionary character, with a broadly solidarity-oriented approach (Jessoula 2019) that facilitated early retirement for specific worker categories.²² The subsequent Di Maio–Salvini reform (2019) was framed in explicit opposition to the Fornero reform and was characterized by a further expansionary trend (Jessoula 2019). Its flagship provision, *Quota 100*, allowed early retirement upon accumulating a combined total of 100 years of age and contribution senior-

²⁰The minimum contributory period for an old-age pension under the Dini system.

²¹For a critical analysis of these measures see Nocella (2013); for a comprehensive enumeration see Camera dei Deputati – Servizio Studi (2021).

²²This was achieved primarily through the *Anticipo Pensionistico* (APE), both in its financial and social variants. The financial APE allowed retirement up to 3 years and 7 months before the statutory age; the social variant was available subject to income and categorical conditions.

ity (minimum 62 years of age plus 38 years of contributions), without any link to demographic developments and applicable until 2021.²³ Since *Quota 100*, no comprehensive structural reform has been implemented; subsequent interventions have addressed indexation (2019–2023) and retirement access through *Quota 102* (2021) and *Quota 103* (2022), without modifying the fundamental architecture of the system.

²³The cumulative fiscal cost was estimated by the Italian Public Accounts Observatory at approximately €30 billion.

3 Japan

Japan's pension system combines a universal flat-rate base pension with an earnings-related component for employees. Successive reforms since 1985 have progressively tightened access conditions, reduced benefit levels, and introduced automatic stabilization mechanisms—all against the backdrop of the world's most advanced demographic aging process.

3.1 Origins of the Public Pension System

The foundations of modern Japanese public pension provision were laid in 1941–1942, with the creation of an earnings-related old-age insurance scheme for employees, initially funded on a fully-funded basis (National Institute of Population and Social Security Research (IPSS) 2014). In 1954, the employee pension was restructured to include both a flat component and an earnings-related component (Takayama 2005). From 1959, coverage was extended to public-sector employees through a network of Mutual Aid Associations (MAA). No provision existed for the self-employed. In 1961, this gap was closed with the introduction of the National Pension (NP), a flat-premium, flat-benefit scheme available after 25 years of participation and at age 65 (Clark 1991). During the 1970s, benefits were expanded substantially: the replacement rate rose from approximately 40% to 60%, aided by the introduction of wage indexation for both schemes in 1973, alongside an increase in the contribution rate (from 3% to 7.8%) (Takayama 2005).

By the early 1980s, the system comprised eight programs covering roughly 90% of the population: the Employees' Pension Insurance (EPI), five other employee programs,²⁴ a seamen's program, and the National Pension. Both the EPI and the NP operated as defined-benefit systems. Benefits under the pre-reform National Pension were calculated as:

$$\text{Monthly benefit} = 1,680 \times \frac{A + \frac{B}{3}}{12} \quad (4)$$

where A is the number of months of paid contributions and B the number of exempt months (up to a maximum of 480 months, i.e. 40 contribution years).²⁵ Benefits under the EPI were calculated as:

$$\text{Monthly benefit} = (2,400 \times N) + (1\% \times R \times N) \quad (5)$$

²⁴Covering central government employees, local government employees, employees of public corporations, private school employees, and agricultural sector employees.

²⁵"A" includes actual contribution months, coverage periods granted as a spouse of an employee, and coverage periods provided by employee pension funds.

where N is the number of contribution years and R the monthly pensionable wage.²⁶ Access to the NP required 20 contribution years and age 60; the EPI required 25 contribution years and age 65.

3.2 The 1985–1986 Reform

The shift toward a more restrictive stance was driven by the financial difficulties of the system at the time, themselves a legacy of the expansionary pension policy of the 1970s (Shinkawa 2003). In particular, the National Pension had come under financial stress due to structural changes in the insured population.²⁷ In this context, a comprehensive reform was implemented between 1985 and 1986.

The reform established a two-tier system providing universal coverage. Under the new architecture, all residents are enrolled in the Base Pension (BP), financed through the National Pension, irrespective of occupation. Insured persons are divided into three categories: (1) the self-employed and other non-employee categories, covered only by the BP; (2) employees, covered by both the BP and the EPI; and (3) spouses of employees, covered by the BP without contributing. The reform also modified benefit formulas (detailed below) and established a quinquennial review of system parameters—retirement age, contribution rate, and the flat NP premium—which was carried out punctually from 1994 to 2004.

3.2.1 The National Pension (Base Pension)

The reformed National Pension replaced the pre-1961 autonomous pension for the self-employed and absorbed the flat component of the employee pension. Workers born before 1926 were grandfathered under the pre-reform rules. For all subsequent cohorts, access to the Base Pension required 25 contribution years and age 65, with actuarial adjustments for early or late retirement relative to that age. Once fully phased in, the monthly benefit would be calculated as:

$$\text{Monthly benefit} = 1,250 \times \frac{A + \frac{B}{3}}{12} \quad (6)$$

The reduction in the benefit coefficient from 1,680 to 1,250 implied lower benefits *ceteris paribus*. The transition was gradual, with the coefficient declining progressively across birth cohorts.²⁸ The Base Pension would be financed by three sources: (a)

²⁶The pensionable wage was determined by 31 income bands. For workers in Band 16 or below (monthly pensionable wage below 160,000 yen), the replacement rate after 40 contribution years was at least 100%, reaching a peak of 181% for the lowest band. The minimum replacement rate, for Band 31 workers, was 60.4%.

²⁷Employment in the agricultural sector fell from 8.42 million in 1970 to 5.02 million in 1982, reducing agriculture's share of the workforce from 16% to 9% (Shinkawa 2003).

²⁸See Clark (1991, p. 37).

flat monthly premiums from the self-employed, indexed to prices; (b) employer contributions for enrolled employees, covering both EPI and BP; and (c) a statutory one-third state subsidy from general tax revenues.

3.2.2 *The Employees' Pension Insurance*

Prior to 1985, EPI benefits comprised a flat component and an earnings-related component (Equation (5)). The reform removed the flat component from the EPI, transferring it to the Base Pension. During the transition, the flat component was phased down gradually from 2,400 yen (the pre-reform EPI flat amount) to 1,250 yen (the BP level). The earnings-related accrual rate was also reduced progressively—from 1.0% to 0.75%—based on the worker's age cohort. The pensionable wage was redefined as the career average of all wages, revalued by EPI-covered wage growth. Access to the EPI required age 60 and 25 contribution years.²⁹ Women employees' retirement age was raised gradually from 55 to 60 across birth cohorts.

3.3 Actuarial Updates (1994–2004) and the 2012 Reform

Consistent with the 1985 framework, several parametric adjustments were made in subsequent years. In 1994, the age at which employees could receive the Base Pension was raised gradually to 65, to eliminate the asymmetry between employees (who could access the BP at 60) and Category 1 insured persons (who could not access it before 65). Full harmonization was completed by 2018.

In 1999,³⁰ two additional measures were introduced for EPI benefits: a direct 5% reduction in benefits for all new entrants from the year 2000³¹ and the revaluation of past contributions at price growth rather than wage growth after age 65. This second measure produces a progressive decline in the real replacement rate over time, estimated at approximately 20% by age 87 (Takayama 2005). The pensionable age for the earnings-related component was raised gradually from 60 (in 2000) to 65 (in 2013, with women following by 2018). The statutory contribution rate was reduced from 17.35% to 13.58%, but the effective contribution increased because bonuses—accounting for roughly 30% of base salary—were incorporated into the contribution base.³² Additionally, the state subsidy to the Base Pension was increased from one-third to one-half of BP expenditure, to be achieved by 2004 (Sakamoto

²⁹The asymmetry between the NP access age (65) and the EPI access age (60) was bridged by a “transitional benefit” paid by the EPI or MAA until the worker reached NP eligibility, effectively providing the Base Pension five years early for employees.

³⁰For further context on this reform, see Sakamoto (2001).

³¹More precisely, the accrual rate was reduced from 7.75% to 7.715%.

³²Denoting R the base wage, the effective contribution moved from $0.1735 \times 1.003R \approx 0.174R$ to $0.1358 \times 1.3R \approx 0.176R$. The larger the bonus share, the greater the effective increase.

2001).³³

As Shinkawa (2003) notes, both the 1994 and 1999 reforms relied on demographic projections that proved overly optimistic. From 2001, the EPI began recording persistent losses, a trend that continued in 2002. Takayama (2005, 2017) emphasizes the difficulty of raising contribution rates further, given the already substantial burden of social contributions and rising dropout rates.

The 2004 reform raised EPI contribution rates from 13.58% to 18.30% by 2017, increasing by 35.4 basis points annually. NP flat premiums rose in parallel. The state subsidy to the BP was increased from 33.3% to 50%, implemented gradually through 2009.

Crucially, the 2004 reform introduced an automatic adjustment mechanism—the *macroeconomic slide*—into the EPI benefit formula.³⁴ Two demographic factors were embedded in the indexation formula: the first tracked the decline in the number of active contributors; the second was set at a fixed -0.3% per year based on 2004 projections of aging through 2025. Takayama (2005) estimated the first factor at approximately -0.6% per year,³⁵ implying a combined annual drag of approximately -0.9% . By law, these factors could not reduce nominal benefit levels; they would only dampen or annul the upward indexation from price or wage growth.³⁶ The mechanism would be deactivated once benefits reached a replacement rate of 50%—projected for 2023. This conditionality limited the automatic character of the adjustment (Sakamoto 2005; Takayama 2010).

No comprehensive reform followed in 2009. The activation of the macroeconomic slide was postponed to 2012, as was the completion of the 50% state subsidy to the BP, pending identification of funding sources. A substantive reform was finally enacted in 2012, financed by a phased increase in the consumption tax: from 5% (2012) to 8% (2014) and 10% (originally 2015, later postponed to 2018),³⁷ with the additional revenues dedicated from 2014 to financing the 50% state subsidy already legislated in 2004. The 2012 reform also merged the Mutual Aid Associations into the EPI from 2015, reducing the minimum contribution requirement for NP eligibility from 25 to 10 years, extended EPI coverage to part-time workers from 2016, removed indexation freezes, and introduced a supplement for low-income retirees (Takayama

³³The increase was delayed to 2009 due to Japan’s economic stagnation, which precluded raising the consumption tax (Sakamoto 2005).

³⁴This mechanism is modeled formally in Kashiwase, Nozaki, and Tokuoka (2012, Box 1).

³⁵Based on a projected decline in EPI participants from 69.4 million in 2005 to 61 million in 2025.

³⁶For example, if prices grew by 0.5% and the demographic drag was -0.9% , the net indexation would be set to zero rather than -0.4% , to avoid a nominal benefit cut. The prevalence of deflation in the 2000s exposed a loophole: in a deflationary environment, the demographic factors would compound an already falling price base, leading to the postponement of the mechanism’s activation until fiscal year 2015 (Takayama 2017; Nomura 2019).

³⁷The increase was earmarked for social expenditure (Takayama 2012).

2012).

3.4 Current Framework: Indexation and Adjustment Rates

The 2012 reform was the last comprehensive revision of the system. Since fiscal year 2013, the regime has operated under the parameters established in 1985 and subsequently modified: premiums follow the price level, while benefits are indexed to wage growth up to age 65 and to price growth thereafter, net of the two demographic adjustment factors. Between 2013 and 2015, benefits were reduced by a cumulative 2.5 percentage points to compensate for the indexation freeze during the deflation of 2000–2002. The annual adjustment rates for the National Pension since 2013 are reported in Table 4.

Table 4. Annual adjustment rates for the Japanese public pension system, 2013–2024. Source: NENKIN (Japan Pension Service), periodic notices.

	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Rate (%)	−1.00	−0.70	0.90	0.00	−0.10	0.10	0.20	−0.10	−0.10	0.40	2.20	2.70

4 Greece

Greece’s pension system entered the 2010s carrying decades of deferred reform, fragmented governance, and financial imbalances financed through general transfers. The three bail-out programs negotiated between 2010 and 2015 compressed into five years a structural overhaul that other countries had spread across three decades. A further reform in 2021 introduced a funded pillar for new entrants.

4.1 Historical Background

Founded in 1934, the main Greek pension institution (IKA) did not undergo regular actuarial reviews for decades. The system fragmented progressively along occupational, benefit-type, and demographic lines, giving rise to what Tinios (2016) describes as a “myopic governance” structure—one that privileged clientelistic advantages and permitted retirement well before age 60 for numerous categories (Petmesidou 2017).

It was not until the late 1970s that the first sustainability concerns emerged (Tinios 2003), followed by a decade of legislative inertia during the 1980s and 1990s (Petmesidou 2017). Growing pension expenditure was financed through ad hoc government transfers rather than increases in contribution revenues, producing uncontrolled expenditure growth without a concurrent strengthening of the system’s contribution base (Nektarios and Tinios 2019). Announced structural reforms were repeatedly delayed.³⁸

By 2010, the public pension system exhibited a financial deficit estimated at 7.3% of GDP. The system comprised numerous funds and regimes, alongside several benefit types.³⁹ The main public pension was a defined-benefit scheme, with benefits calculated on the basis of the best five of the last ten contributory years, multiplied by an accrual rate that varied by occupation (ranging from 2% to 3%) and by the number of contribution years. Supplementary public DB funds also existed.

4.2 Reforms during the Crisis and the Bail-Out Programs: 2010, 2012, and 2016

The intervention on the pension system, long overdue, was finally triggered in 2010 in the context of the First Memorandum of Understanding (MoU),⁴⁰ coinciding with the first acute phase of the Greek sovereign debt crisis (Nektarios and Tinios 2019).

³⁸For a detailed account of the failed reform attempts, see Tinios (2003).

³⁹In addition to the main “principal” pension, the system included auxiliary pensions, lump-sum severance benefits, fund dividends, and social pensions.

⁴⁰Between 2010 and 2015, Greece subscribed to three MoUs—in 2010, 2012, and 2015—with the “Troika” of creditors formed by the European Commission, the European Central Bank, and the International Monetary Fund.

The 2010 reform operated on two levels. On the expenditure side, it tightened access to retirement. On the structural side, it redesigned the architecture of the Greek pension system, harmonizing the institutional fragmentation that had accumulated over decades (Symeonidis 2016; Kourachanis 2021).

Early retirement now required age 60; the mandatory old-age retirement age was raised to 65; and the minimum contribution period was raised from 35 to 40 years. Actuarial penalties were introduced for retirement below the statutory age.⁴¹ On the structural side, the reform introduced a two-component benefit design: a flat-rate basic pension (a “Pillar Zero”⁴²) financed from general revenues, alongside an earnings-related component based on a broader wage average and a lower accrual rate (0.8%–1.5%, compared with the pre-reform range of 2%–3%). The reform also mandated the consolidation of all public pension funds by 2015. However, the slow pace of implementation triggered a wave of early retirements before the new rules took effect (Kim, Kangur, and Kalavrezou 2021), limiting the short-run impact and prompting the characterization of the outcome as a “half a reform” (Tinios 2016, p. 80).

Further restrictive measures were implemented in 2012–2013 in the context of the Second MoU. The early retirement age was raised to 62 and the statutory retirement age to 67. Thirteenth and fourteenth month pension payments were first reduced and then eliminated entirely. Indexation was frozen to achieve short-term savings,⁴³ and a “zero-deficit rule” was imposed on the supplementary funds, which had accumulated persistent deficits requiring state intervention. The main supplementary funds were merged into a single entity, ETEA (Kouskouna and Strantzalou 2021).

At the end of the two reform cycles tied to the first two MoUs, the Greek pension system comprised: (a) the basic pension introduced in 2010; (b) an earnings-related pension disbursed by eight separate funds,⁴⁴ each with its own rules;⁴⁵ and (c) various supplementary pensions and transfers. The supplementary funds were merged into ETEA in 2012 and converted to an NDC architecture from 2014, on a pro-rata basis for contribution years after 2014.

Renewed concerns about Greek sovereign debt led to a Third MoU and a further cycle of restrictive reforms, including a pension measure in 2016 (Symeonidis 2016). The 2016 reform pursued two objectives: eliminating remaining inter-regime dispari-

⁴¹A 6% reduction per year of early retirement relative to the standard retirement age.

⁴²A non-contributory pension pillar financed from general tax revenues, following the classification in Holzmann, Hinz, and Dorfman (2008).

⁴³These short-term benefit cuts were subsequently ruled unconstitutional in 2015 (Kim, Kangur, and Kalavrezou 2021).

⁴⁴The funds covered: private-sector salaried employees (with sub-funds for banks, the public electricity company, and other sectors), civil servants, the self-employed, the agricultural sector, professionals (lawyers, physicians, engineers), journalists, seamen, and Bank of Greece employees.

⁴⁵The consolidation mandated in 2010 was never fully implemented.

ties, and reducing short-and medium-term costs by intervening on benefit levels and retirement conditions in the supplementary funds.

Benefits in the earnings-related second tier were reduced through a lower accrual rate (0.77%–2.00%, compared with the post-2010 range).⁴⁶ New retirees were subject to the new formula immediately; existing pensioners were transitioned under specific rules.⁴⁷ The reform imposed convergence of retirement ages toward 67 by 2022 for all cohorts previously exempted, and raised the actuarial penalty for early retirement from 6% to 16% per year below the statutory age. However, workers who had already met eligibility conditions under pre-2015 law were excluded from the new rules, limiting the scope of the reform. Additionally, benefits at the top of the distribution were reduced by applying a 0.45% accrual rate to contributions made before 2014 (which had been subject to the DB formula), while post-2014 contributions remained under the NDC rules already in force since 2012.

4.3 After 2016

In 2017, a further package of measures was proposed, including the universal application of the 2016 cuts (eliminating the transitional exemptions noted above) and a freeze on inflation indexation. This package was never enacted, in part due to the broader climate of opposition to austerity, which was reinforced by a 2019 ruling of the Greek Council of State abrogating the 2014 and 2016 reductions in benefits and the harmonization of regimes. At the end of the three-MoU reform cycle, the system exhibited a three-pillar structure, summarized in Table 5.

Table 5. Structure of the Greek public pension system before the 2021 reform. Source: adapted from Altiparmakov (2023).

Tier	Type	Financing and rules
I	Universal flat-rate basic pension	Financed from general tax revenues.
II	Earnings-related DB pension	Financed by contributions (20%). Accrual rate 1%. State transfers possible.
III	Supplementary NDC pension	Financed by contributions (6%). Autonomous financial balance.

Despite the zero-deficit rule and the NDC architecture, the third tier continued

⁴⁶The accrual rate was increasing in contribution years; see Kouskouna and Strantzalou (2021, Table 4.1).

⁴⁷Where the new formula yielded a lower benefit than the pre-2014 frozen level, benefits were kept at the 2014 level without indexation. Where the new formula yielded a higher benefit, it was applied with indexation.

to experience financial difficulties, as evidenced by successive contribution increases (2015) and benefit cuts (2016).⁴⁸

By 2019, Nektarios and Tinios (2019) characterized the system as: (a) excessively generous, with replacement rates among the highest in the EU for low-income workers; (b) too costly, with high non-wage labor costs that would continue to grow, undermining competitiveness; (c) insufficiently flexible, offering little scope for individual or sectoral differentiation and forcing groups such as the self-employed to over-insure; (d) overdependent on the public pillar, suppressing both the demand for private insurance (given high replacement rates) and its supply (given high social security contribution rates of 27%–38%); and (e) insufficiently transparent, with chronic contribution increases unaccompanied by organic reform, eroding confidence in the system.

4.4 The 2021 Reform

A change in the political environment led to the 2021 reform (Altiparmakov 2023), formally titled “Insurance Reform for the New Generation: Introduction of a Funded System of Defined Contributions in Supplementary Insurance.”⁴⁹

⁴⁸Nektarios and Tinios (2019, p. 17) describe the third tier’s condition as “pay-while-you-can, or even pay-what-you-grab.”

5 Germany

Germany's pension system has moved over three decades from a generous, wage-indexed defined-benefit architecture toward a hybrid arrangement that incorporates automatic stabilization mechanisms and subsidized private supplementary provision. The reform trajectory has been incremental but cumulatively significant.

5.1 Historical Background

The foundations of the modern German pension system date to 1957, when the previously funded scheme was converted into a PAYG system to provide retirement income in a context of severe postwar capital scarcity (Rürup 2002). The 1957 system was designed to maintain workers' living standards in retirement, with contributions calculated as a percentage of gross earnings and benefits linked to contributions accumulated over the full career. No minimum pension guarantee was provided. Benefits were calculated according to:

$$\text{Annual benefit} = P \times N \times PV \quad (7)$$

where P denotes the pension points earned over the working life (proportional to the ratio of the worker's annual wage to the average wage in each year),⁵⁰ N the number of contribution years, and PV the indexation factor. The mandatory retirement age was set at 65, with 60 for women and older unemployed workers.

In 1972, a minimum pension was introduced by establishing a floor on pension points (0.75 per year) (Börsch-Supan and Reil-Held 2001), and an early retirement window between ages 60 and 65 was opened, effectively abolishing mandatory retirement. The 1972 reform's generosity made the German system one of the most generous in the world, projecting an equilibrium contribution rate of 40% by 2035 in light of Germany's demographic trajectory—not dissimilar from that of Italy and Japan—and thus prompted further reform (Börsch-Supan and Wilke 2003).

5.2 The 1992 and 1999 Reforms

The 1992 reform modified the indexation regime established in 1957, which had linked benefits to gross wages (Rüb and Lamping 2010). Under the new rules, pension benefits were indexed to *net* wages rather than gross wages. An actuarial adjustment factor, A , was introduced into the benefit formula based on the age of retirement:

⁵⁰A worker whose wage exactly matched the economy-wide average would accumulate one pension point per year. N is the number of contribution years, and PV the indexation factor. Under the 1957 legislation, if wages grew by 1% in a given year, PV would be 1.01.

$$\text{Annual benefit} = P \times N \times PV \times A \quad (8)$$

The age of 65 was established as the pivot age, with actuarial bonuses (maluses) for deferred (early) retirement.⁵¹

While the 1992 reform aimed to stabilize finances, it did not fully account for the disruptive effects of German reunification on labor market conditions across the east-west divide (Rürup 2002). A supplementary intervention in 1999 introduced a “demographic factor” that would have embedded life expectancy into the benefit formula.⁵² However, this element of the 1999 reform was revoked following the change of government in 1998 (Börsch-Supan and Wilke 2003).

5.3 The Riester Reform (2001) and the Rürup Commission (2004–2007)

The 2001 Riester reform pursued two simultaneous objectives: containing the statutory contribution rate below 22% by 2030,⁵³ and maintaining the public system’s role as a social safety net by targeting a gross replacement rate of 67% in 2030—equivalent to a net replacement rate of 48% in 2020 and 43% in 2030 (Wilke 2009).

The reform modified the indexation formula substantially.⁵⁴ To compensate for the expected reduction in public pension levels, voluntary private supplementary provision was incentivized through the *Riester-Rente* schemes—subsidized defined-contribution funds governed by common regulatory standards and a state supervisory entity—though private provision was never made mandatory (Rüb and Lamping 2010).⁵⁵ Contribution rates to these plans were phased in progressively, reaching 4% of gross earnings by 2008.⁵⁶

It quickly became apparent that the Riester measures would not suffice (Wilke 2009), and in 2002 the Ministry of Health and Social Affairs appointed the Rürup Commission to develop further stabilization proposals. Two sets of proposals were

⁵¹Each year of early retirement entails a 3.6% benefit reduction; each year of delayed retirement increases the pension by 5% (Wilke 2009, p. 49).

⁵²The formula remained identical to Equation (8), with a demographic factor substituted for A: Demographic Factor = $\frac{\frac{LE_{t-2}}{LE_{t-1}} - 1}{2} + 1$. A 1% increase in life expectancy would, *ceteris paribus*, reduce benefits by approximately 0.5%.

⁵³Comparable to the equilibrium contribution rate in PAYG systems.

⁵⁴The new indexation factor PV would be: (a) positively linked to price growth; (b) positively linked to a fixed factor that would decline over time; (c) negatively linked to the contribution rate to private pension funds, which would rise progressively; and (d) negatively linked to the public system contribution rate (Börsch-Supan and Wilke 2003). Formally: $PV = PV_{t-1} \times \left(\frac{R_{t-1}}{R_{t-2}} \times \frac{\frac{d_t}{100} - c_{f,t-1} - \tau_{t-1}}{\frac{d_t}{100} - c_{f,t-2} - \tau_{t-2}} \right)$, where c_f is the private fund contribution rate, τ the public contribution rate, and d_t an adjustment factor.

⁵⁵Eligible fund rules included: disbursement only after age 60; minimum return of contributed capital guaranteed; sex neutrality; lifetime annuity guarantee; transparency obligations regarding portfolio performance, fund structure, and risks; and protection from creditor seizure.

⁵⁶The schedule was: 1% from 2002, 2% from 2004, 3% from 2006, 4% from 2008.

adopted: in 2004, regarding a modified indexation mechanism, and in 2007, regarding the statutory retirement age.

In 2004, the demographic factor revoked after 1998 was replaced by a *sustainability factor*: indexation to average gross wages would be weighted by growth in the system dependency ratio, defined as the ratio of the number of pension recipients to the number of active contributors.⁵⁷ This introduced the first genuine automatic self-adjustment mechanism into the German system. According to Rüb and Lamping (2010), its main consequences were: (a) linking pension levels to aggregate demographic developments; (b) making benefit levels sensitive to labor force size; and (c) ensuring that growing private fund contributions would automatically reduce public pension levels.

Alongside these measures, the third pillar was administratively reformed, and an equivalent scheme for the self-employed—the *Rürup-Rente*—was introduced from 2005.⁵⁸

In 2007, the statutory retirement age was raised: from 65 to 66 between 2012 and 2029 at a rate of one month per year, and from 66 to 67 between 2024 and 2029 at a rate of two months per year. The eligibility age for disability pensions was raised from 63 to 65 between 2017 and 2029, and the early retirement age (with actuarial deductions) from 60 to 62. A pathway allowing retirement at age 65 with 45 contribution years and no actuarial deductions was retained. The literature notes that this retirement age increase was “discretionary” and not linked to life expectancy—which would continue to grow after 2030—potentially compromising the long-run achievement of the 43% net replacement rate target (Vogt and Althammer 2016).⁵⁹

The retirement access conditions resulting from the sequence of reforms in 1992, 1999, and 2007 are illustrated in Figure 6, which shows the statutory ages applicable to different retirement pathways.

⁵⁷The adjustment factor took the form: $\left(1 - \frac{IDS_{t-2}}{IDS_{t-3}}\right) \times \alpha + 1$, with $\alpha = 0.25$. See also Vogt and Althammer (2016).

⁵⁸For a performance analysis of both Riester and Rürup funds, see Mulder, de Khoo, and Zettelmeyer (2022).

⁵⁹The scenario analysis in Vogt and Althammer (2016) finds that a statutory retirement age fixed at 67 from 2030 performs markedly worse on a long horizon than scenarios linking the retirement age to life expectancy.

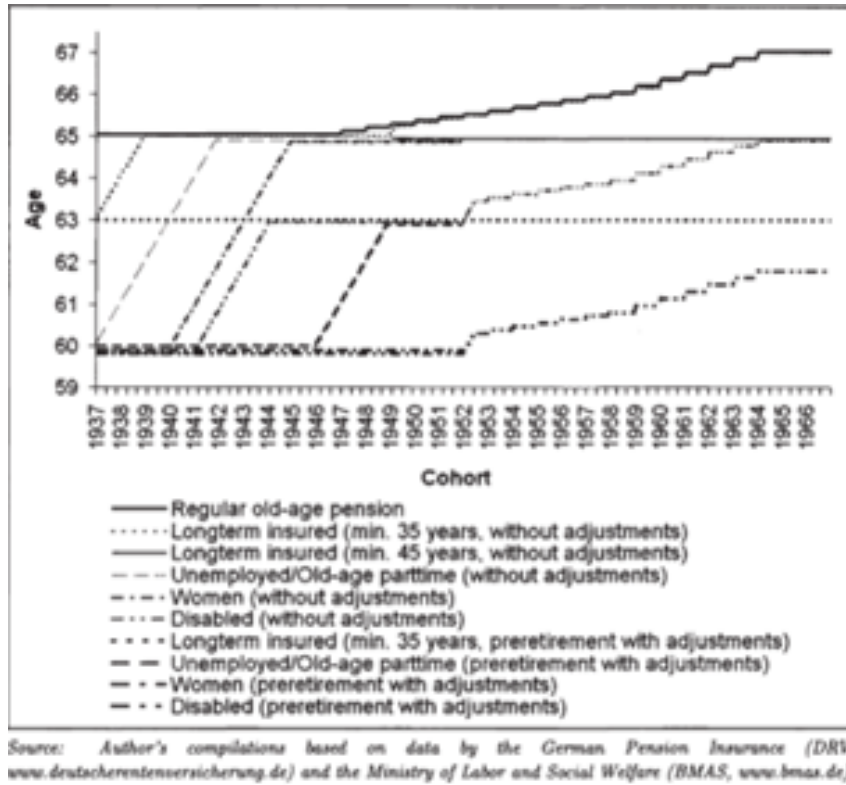


Figure 6. Changes in retirement ages following the 1999, 2001, and 2007 reforms. Source: Wilke (2009, p. 74).

Rüb and Lamping (2010) characterizes the outcome of this reform sequence as a hybrid system that had incorporated several features typically associated with DC architectures,⁶⁰ having abandoned many of the Bismarckian principles on which the system was founded.⁶¹

5.4 After 2007

Following the 2007 reform, and partly in response to the global financial crisis, the restrictive reform trajectory was interrupted. Subsequent interventions have been broadly expansionary.

In 2009, the *Schutzklausel* (“Protection Clause”) was introduced to prevent deflation from triggering nominal benefit cuts (Guardiancich and Natali 2021). In 2014, the minimum retirement age for workers with at least 45 contribution years was reduced from 65 to 63, and the eligibility criteria were broadened to include periods

⁶⁰Both Bonin (2009) and Börsch-Supan and Wilke (2003) describe the sustainability factor introduced in 2004 as an automatic mechanism characteristic of NDC systems, while Rüb and Lamping (2010) classify the system overall as “hybrid,” given the DB structure of the public component alongside the DC structure of the private one.

⁶¹Rüb and Lamping (2010) note the transition from a goal of maintaining living standards to one of providing basic income security, alongside the shift toward greater weight on individual saving decisions through the *Riester-Rente*.

of unemployment and voluntary contributions, increasing the number of potential beneficiaries.

The *Flexirentengesetz* (Flexible Pension Reform Act), enacted in December 2016, introduced new incentives to extend working lives and measures facilitating a gradual transition from work to retirement, including the relaxation of rules governing the simultaneous receipt of wage and pension income (Guardiancich and Natali 2021).

The *Rentenpaket* of 2018 committed the government to maintain the standard gross replacement rate at a minimum of 48% through 2025 (European Policy Committee and Social Policy Committee 2020). A year later, legislation was approved for a basic pension (*Grundrente*) for long-term contributors, enacted from 2021. Workers with at least 33 contribution years⁶² whose career earnings fell below 80% of the average wage are eligible for a means-tested supplement.

⁶²Qualifying periods include childcare (up to 10 years per child), family care, and unemployment with benefit receipt.

6 France

France has consistently resisted structural pension reform, relying instead on repeated parametric adjustments to a system characterized by institutional complexity and multiple regimes. The central lever has been the progressive extension of the required contributory period, alongside modest increases in the statutory retirement age. Despite several attempts at systemic overhaul, every reform since 1993 has remained parametric in nature.

6.1 Historical Background

6.1.1 *The Creation of the System and the Expansionary Phase*

The origins of the French pension system can be traced to the mid-nineteenth century, with the granting of legal recognition to Mutual Benefit Societies and the creation of a mandatory pension scheme for public employees.⁶³ In the early twentieth century, a compulsory pension regime was established (1910), followed by a funded pension scheme for workers in state-owned industries (1930s). A PAYG system was not introduced until 1941; after the war, a unified compulsory public pension system was established in 1945, with benefits available at age 65 after 30 years of contributions. Participants in “special” regimes⁶⁴ were permitted to retain their own arrangements. The 1960s, 1970s, and 1980s were an expansionary period, during which early retirement was frequently used as an instrument of labor market policy (Guillemard 2020).

By the early 1990s, the future sustainability challenges of the French system were becoming apparent, as reflected in the White Paper on Pensions (Rocard and Commissariat Général du Plan 1991) and in the broader policy debate synthesized in Blanchet and Legros (2002). This debate, in the wider European context of pension reforms,⁶⁵ culminated in the Balladur reform of 1993.

6.1.2 *The French Pension System in the Early 1990s*

Architecture. ⁶⁶ The first pillar is public and mandatory, financed on a PAYG basis through contributions split between workers and employers. It consists of

⁶³This section is primarily based on Secrétariat général du Conseil d’orientation des retraites (2022).

⁶⁴*Régimes spéciaux* are schemes covering specific occupational categories whose rules diverge from the general regime, such as that of the Banque de France.

⁶⁵Including the Amato reform in Italy (1992) and the German reform of the same year.

⁶⁶The French system has always been and remains highly complex, with numerous types of benefits and regimes operating under widely divergent rules. For clarity, the description focuses on the system’s current characteristics, given the predominantly parametric nature of the reforms applied in France (OECD 2023).

the *Régime Général*⁶⁷ for employees, and the *Régime Social des Indépendants* (RSI) for the self-employed, itself subdivided into sector-specific schemes referred to as “aligned” regimes because they operate under rules broadly harmonized with the *Régime Général*.

The second pillar is managed by social partners, with each trade union maintaining its own regime; enrollment is mandatory for all employees. In general terms, the second pillar is also PAYG-financed, with benefits calculated using a points system: accumulated points, multiplied by a reference wage value, determine the pension (Guillemard 2020, p. 5). Over time, second-pillar schemes were consolidated under AGIRC (for managerial staff) and ARRCO (for non-managerial employees),⁶⁸ which merged in 2019. The third pillar consists of voluntary, privately managed individual schemes.

Cutting across the first and second pillars were the “integrated regimes” for public employees, which covered both basic and supplementary pension functions under rules differing substantially from those of the private sector. Additionally, the “special regimes” applied to specific occupational categories and could diverge significantly from the general framework.

Benefit formula. ⁶⁹ First-pillar benefits are calculated as:

$$\text{Benefit} = \beta \times SAM \times \frac{N}{N_{\max}} \quad (9)$$

where β is the accrual rate (*taux plein*), SAM is the reference wage (mean of indexed past wages), and $N/N_{\max}$ is the *proratisation* coefficient expressing the worker’s contributory years relative to the statutory maximum. A worker is entitled to the full rate β (*taux plein*) by either: (a) reaching the *âge du taux maximum*—the upper age threshold, which requires no minimum contributory period; or (b) reaching the *âge d’ouverture des droits* (AOD, the statutory retirement age), a lower threshold, conditional on satisfying the required contributory period. Retirement before either threshold is possible but incurs an actuarial reduction; delayed retirement beyond the threshold has historically generated a supplement.

As of 1993, the parameters were: $\beta = 0.50$; *âge du taux maximum* = 65; AOD = 60; SAM computed as the average of the best ten contributory years; maximum statutory contributory period ($N_{\max}$) = 37.5 years (i.e., 150 quarters).

⁶⁷The nomenclature of French institutions is retained in the original French to preserve meaning.

⁶⁸“Association Générale des Institutions de Retraite Complémentaire des Cadres” and “Association pour le Régime de Retraite Complémentaire des Salariés,” merged in 2019.

⁶⁹This subsection draws on Kasagi (2017).

6.2 The Balladur Reform (1993)

The 1993 Balladur reform⁷⁰ applied exclusively to the *Régime Général* and the aligned regimes, leaving the public-sector integrated regimes and the second pillar unchanged.

The reform’s two principal axes were an increase in the required contributory period and a reduction in benefit generosity. On the former, the minimum contributory period for a full-rate pension (*au taux plein*) was extended from 37.5 to 40 years (150 to 160 quarters),⁷¹ implemented gradually between 1994 and 2003.

The reform thus set two pathways to the full rate: retirement at the *âge du taux maximum* (65), with no contributory condition, or retirement at the AOD (60 in 1993), conditional on meeting the 40-year requirement. Retirement at the AOD before completing the required period remained possible, but each missing quarter (six-month period) reduced the benefit by 2.5%, up to a maximum reduction of 10%. On the benefit computation side, the *SAM* base was broadened from the best ten to the best twenty-five contributory years, phased in at a rate of one additional year per year from 1993 to 2010.⁷² Indexation of past wages and the revaluation of current pensions were also switched from wage growth to price growth, producing a further reduction in the real level of future benefits.

6.3 Parametric Reforms: Fillon (2003), Woerth (2010), and Touraine (2014)

Ten years after Balladur, the 2003 Fillon reform⁷³ harmonized retirement conditions between the private and public sectors. The required contributory period for civil servants was aligned with the private-sector rules by 2008. For private-sector workers, the required contributory period for the full rate was extended again—from 40 to 41 years by 2012, with a corresponding adjustment to $N_{\max}$. After 2012, the government was required to set new contributory period requirements every four years, as a function of life expectancy.

The reform introduced special provisions for workers with “long careers” who had entered the labor market very young.⁷⁴ Bonus and malus incentives were

⁷⁰This section draws on Lavigne (2014).

⁷¹The reform referred exclusively to quarters. The phase-in ran from 1994 to 2003, with the requirement rising by one quarter per year.

⁷²The use of the best 25 years, revalued at price inflation, yields a lower base than the best 10 years revalued at wage growth. Over the long run, this could reduce pensioners’ living standards by up to 20% relative to the pre-reform system, returning them to 1980s levels (80–90% of active income), according to the French Pension Advisory Council (Blanchet 2020).

⁷³This section draws on Jeger and Lelièvre (2005).

⁷⁴Retirement was possible after 42 contribution years and at age 56, a measure that perversely worsened the short-term fiscal balance by allowing 250,000 workers to retire before age 60 between 2004 and 2005 (Jeger and Lelièvre 2005).

introduced to encourage continued work beyond the standard retirement threshold.⁷⁵ The mandatory retirement age—the age at which employers could force retirement without worker consent—was raised from 60 to 65.⁷⁶ Price indexation of past wages and pension revaluation was extended to the integrated regimes of public employees, and the special regimes were aligned by raising their required contributory period from 35 to 40 years.

In 2010, in the context of the global financial crisis, the Woerth reform aimed to reduce pension fund deficits (Guillemard 2020). The *âge du taux maximum* was raised from 65 to 67, and the AOD from 60 to 62, for both private-sector workers (general and aligned regimes) and public-sector workers (integrated regimes). Workers in arduous occupations and those with disabilities were exempted. Entitlement conditions for civil servants—including provisions allowing retirement after 15 years of service with three or more children—were tightened. The civil servants’ accrual rate (β) was reduced concurrently (Blanchet 2020).

The 2014 Touraine reform, originally conceived as a systemic reform but ultimately remaining parametric (Guillemard 2020), applied only to the general and aligned regimes. The required contributory period was raised by three quarters every three years, reaching 43 years for workers born from 1973. Contribution rates were also increased gradually by a cumulative 0.6 percentage points over four years. Additional protections for arduous occupations were introduced alongside administrative simplifications.⁷⁷

Despite three successive attempts at parametric reform (2003, 2010, and 2014), systemic ambitions repeatedly went unrealized. The Borne reform of 2023 encountered the same fate.

6.4 The Borne Reform (2023)

Pension reform was a central commitment of the 2017 presidential campaign, but the legislative process⁷⁸ abandoned its original systemic design, partly due to the withdrawal of the first draft in February 2020 following the outbreak of the Covid-19 pandemic. The version ultimately enacted—subsequently termed the “Borne reform”—was filed with the National Assembly in January 2023 and entered into force in September of that year. Of the original reform project, three main provisions were retained, explicitly aimed at reducing costs by 2030 and harmonizing the system

⁷⁵A 0.75% bonus per additional quarter worked (3% per year), up to a maximum of 12 quarters (+9%); symmetrically, the malus for early retirement was reduced from 2.5% per quarter under Balladur to 1.25% per missing quarter. In 2008, the bonus was raised to 1.25% per additional semester.

⁷⁶Raised further to 70 in 2008.

⁷⁷See Lavigne (2014, pp. 26–29).

⁷⁸For the development of the reform bill between 2017 and 2020, see Guillemard (2020, pp. 11–23); for the architecture of the original proposal, see Boulhol (2020).

(Bavaro and D’Onghia 2023; Gannon, Legros, and Touzé 2023).

The most significant—and most contested—measure was the gradual increase in the AOD from 62 to 64. A second measure accelerated the increase in the required contributory period to 43 years, as already planned under the Touraine reform (Figure 7).

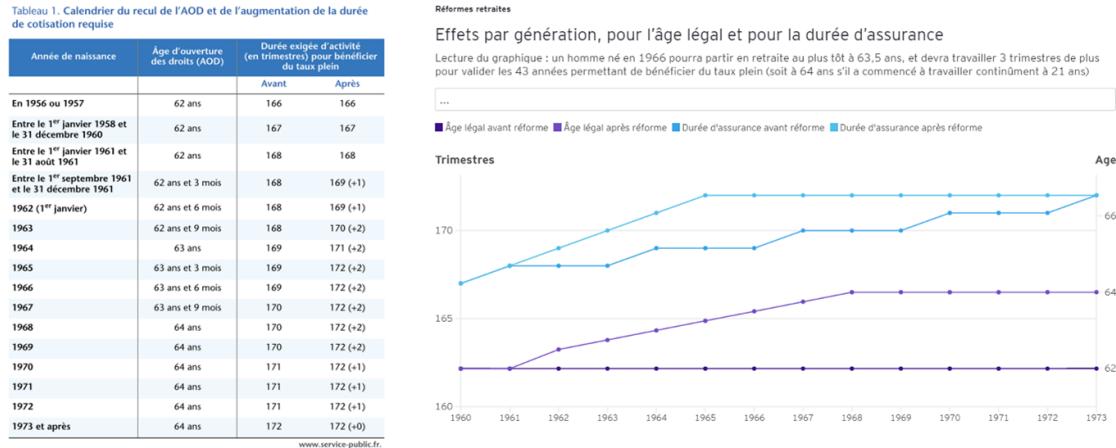


Figure 7. Increases in the *âge du taux maximum* and required contributory period under the Borne reform (left panel); generational effects of the reform (right panel). Source: Gannon, Legros, and Touzé (2023, p. 21) (left); EY (2024) (right).

The reform created asymmetric effects across cohorts. Workers with long careers—those who started early and accumulated 43 contribution years well before age 64—were disproportionately penalized. A worker who began employment at 18 would have completed the 43-year requirement at 61. Before the reform, such a worker could have retired at 62 with 44 years of contributions; after the reform, retirement under equivalent conditions would require working until 64, with 45 years of contributions (Gannon, Legros, and Touzé 2023).

The reform was also intended to incentivize employment beyond age 65; however, it generated asymmetric outcomes between workers in arduous occupations and those in more physically sustainable roles (Bavaro and D’Onghia 2023), and created difficulties for workers unlikely to maintain employment after age 60 (Gannon, Legros, and Touzé 2023).

A third provision mandated the harmonization of most “special regimes,” which had historically operated under more favorable retirement conditions than the general regime. As shown in Figure 8, only 10% of special-regime enrollees retired after age 62, compared with 25% of civil servants subject to the general regime, 30% of private-sector employees, and 45% of the self-employed.

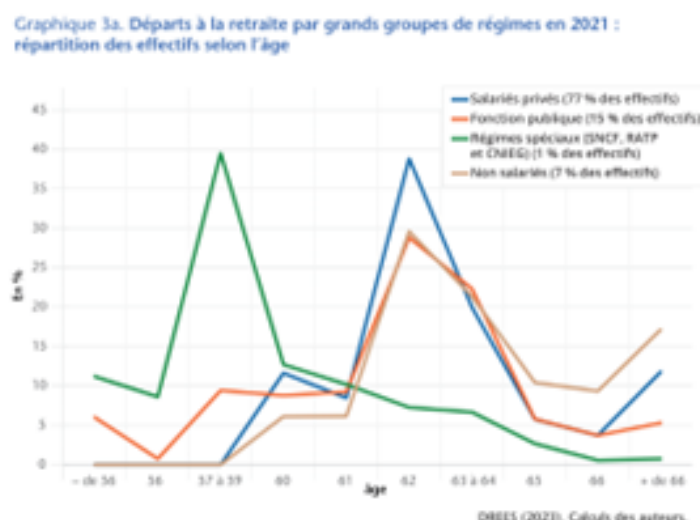


Figure 8. Distribution of effective retirement age by group in 2021. Source: Gannon, Legros, and Touzé (2023, p. 33).

The evolution of French pension parameters across the main reforms is summarized in Figure 9, which illustrates the gradual but consistent upward trend in both the statutory retirement age and the required contributory period, with a corresponding increase in effective retirement ages over time.

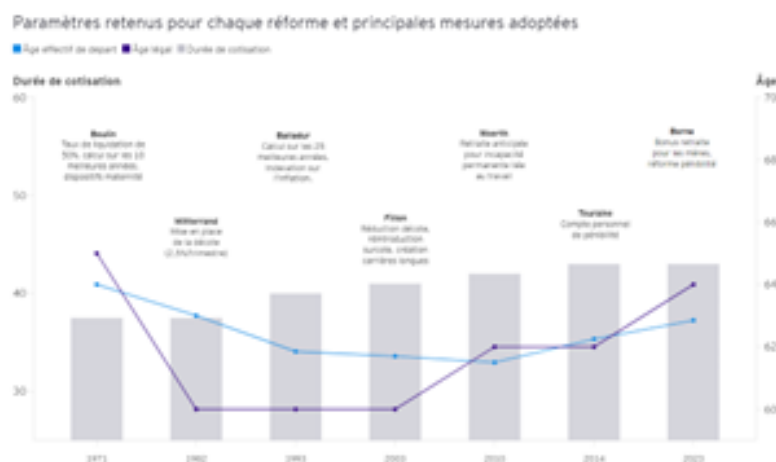


Figure 9. Key parameters of the French pension system across reforms. Source: EY (2024).

7 A Taxonomy of Pension Reform Instruments

The cross-country survey reveals a set of recurring policy levers activated by governments in response to demographic aging. This section organizes these interventions into a taxonomy, drawing on the classification framework proposed in the European Commission’s Ageing Report (European Commission 2023) and elaborated in Tavernier and Boulhol (2021).⁷⁹

I. Structural versus parametric reforms. All five countries have implemented parametric reforms⁸⁰—modifications to retirement eligibility and benefit computation rules within an unchanged overall system architecture. France is the most consistent case: despite several projects for systemic overhaul (in 2014 and again during 2017–2020), every reform enacted has remained parametric, involving incremental changes to the contributory period and, more recently, the statutory retirement age.

The other four countries also undertook structural reforms, broadly in line with the prescriptions of international institutions.⁸¹ Italy introduced an NDC system in 1995, with a gradual phaseout of the pre-existing DB architecture. Japan harmonized and consolidated its multiple regimes and established the current two-tier structure in 1985. Greece redesigned its system in 2010 and introduced NDC for the earnings-related component of the public pension in 2012. Germany incorporated sustainability and contribution factors between 1992 and 2004, moving the system toward a hybrid DB/DC architecture.

II. Increases in statutory and contributory retirement ages. All systems have raised either the statutory retirement age, the required contributory period, or both (European Commission 2023). Some countries have employed automatic mechanisms linking these parameters to life expectancy (Italy and Greece, discussed under item V below); others—Germany and France—have relied on discretionary parametric increases.⁸² Looking ahead, the statutory retirement age is projected to rise by five years in Greece by 2070, by two years in France by 2050, by one year in Germany, and by one year in Japan.

Alongside increases in formal retirement ages, all five countries have implemented measures designed to discourage early retirement and incentivize continued employ-

⁷⁹For additional background on automatic adjustment mechanisms, see also Vidal-Meliá, Boado-Penas, and Settergren (2009).

⁸⁰A parametric reform is defined as one that modifies retirement eligibility conditions without altering the fundamental architecture or rationale of the system (OECD 2023).

⁸¹For example, World Bank (1994).

⁸²The term “discretionary” reflects the absence of a direct link to life expectancy: adjustments are decided periodically by the policymaker rather than triggered automatically by a demographic rule.

ment. These include the closure of early retirement windows (Greece 2010 and 2014; Japan 1985; Germany 1992; France 2003) and the introduction of actuarial bonuses for deferral beyond the standard eligibility threshold.

III. Reductions in benefit generosity. A third category encompasses measures that reduce the level of benefits through modifications to benefit formulas, the definition of pensionable wages, and indexation rules.

In DB systems where the pension is proportional to the average wage over a reference period, reforms have typically extended the reference period to include more years—compressing the average toward the career mean—or reduced the accrual rate directly. Reference period extensions were implemented in Japan (1985), Italy (1992), France (1993), and Greece (2010); accrual rate reductions were enacted in Japan (1985 and 1999), Greece (2010, 2014, and 2016), and France (2010).

In Germany, where the system uses a points-based formula, reforms have focused on the indexation of pension values: switching from gross to net wages (1992), introducing a demographic weighting into price indexation (2001), and converting this weighting into an automatic sustainability factor (2004) that reduces benefit growth when the system dependency ratio deteriorates. Japan implemented a partial indexation adjustment from 2004, weighted by demographic factors, though without a fully automatic mechanism. Italy (2011) and France (2023) introduced partial de-indexation for higher pensions. Emergency indexation freezes were imposed during periods of acute crisis in Greece (2015) and Italy (2011).

IV. Increases in contribution revenues. Alongside expenditure-side measures, all countries have also intervened on the revenue side. France and Japan raised statutory contribution rates. Germany and Japan increased the share of general tax revenues financing pension expenditure, with Japan raising the statutory state subsidy to the Base Pension from one-third to one-half of expenditure. Several countries also established reserve funds to buffer the fiscal costs of the demographic transition, a notable case being France’s *Fonds de Réserve pour les Retraites*.

V. Automatic adjustment mechanisms. The fifth category encompasses mechanisms that adjust pension system parameters automatically in response to demographic or macroeconomic developments, without requiring repeated legislative action. Following the classification in European Commission (2023) and Tavernier and Boulhol (2021), three sub-types can be distinguished: (a) automatic balancing mechanisms, which reduce indexation to prevent pension fund deficits; (b) sustainability factors, i.e. benefit computation coefficients that decline with life expectancy; and (c) automatic links between the statutory retirement age and life expectancy.

Germany’s evolving indexation formula—modified in 2001 and made fully automatic in 2004 through the sustainability factor—represents a case of sub-type (a), as does Japan’s macroeconomic slide introduced in 2004. The NDC transformation coefficients in Italy (from 1995) and Greece (from 2012) represent sub-type (b): longer life expectancy automatically reduces the annuity conversion factor and hence the annual benefit, preserving aggregate pension expenditure. Sub-type (c) is represented by the life-expectancy linkage of Italy’s statutory retirement age under the Fornero reform (2011) and by the French *proratisation* coefficient, whose denominator ($N_{\max}$) is reviewed in line with projected retirement age, effectively embedding a demographic update into the benefit formula from 2003 onward.

8 Conclusions

This paper has examined the principal policy levers used to reform the public first pillar of pension provision in five aging economies—Italy, Japan, Greece, Germany, and France—from the 1990s onward. The selection of these countries was motivated by their shared demographic pressures and the dominant role of public PAYG systems in their retirement income architecture.

The analysis proceeded in two steps. First, Chapter 1 documented the mechanics of PAYG systems and established their link to demographic dynamics, before characterizing the evolution and projected trajectory of key demographic indicators for each country. The data confirm that all five countries face sustained demographic aging, driven by below-replacement fertility and rising longevity—though the intensity and timing of this process differ, with Japan, Italy, and Greece already experiencing its sharpest effects, while Germany and France face a slower and later onset.

Second, Chapters 2 through 6 surveyed the major restrictive interventions country by country, situating each reform in its political and economic context, describing the changes enacted, and—where relevant—reviewing the academic literature on their social consequences.

The taxonomy constructed in Chapter 7 identifies five categories of reform instruments common across the five countries. *Parametric reforms* are universal: all five countries have raised retirement ages and contributory periods, often repeatedly. France stands apart as the only country that has never moved beyond parametric reform despite several systemic reform projects. The other four have also undertaken *structural reforms*—most notably the shift from defined-benefit to notional defined-contribution architectures—in line with prescriptions from international institutions.

Reductions in benefit generosity have been achieved through a combination of formula modifications, extended pensionable wage reference periods, reduced accrual rates, and weaker indexation. Emergency indexation freezes have also been used during periods of acute fiscal stress. On the revenue side, all countries have increased *contribution rates or general-revenue transfers* to pension systems, in some cases establishing dedicated reserve funds.

Perhaps the most consequential development across these five cases is the growing reliance on *automatic adjustment mechanisms*—whether automatic balancing rules, sustainability factors, or retirement-age links to life expectancy. These mechanisms represent a qualitative shift in reform strategy: rather than legislating fixed parameters that require repeated renegotiation as demographics change, they embed demographic adaptation directly into the system’s operating rules, reducing political transaction costs and improving long-run fiscal predictability.

Taken together, the evidence suggests a convergent but path-dependent reform

trajectory. Countries with stronger fiscal constraints, more acute demographic pressures, or crisis-driven political openings have tended to implement more rapid and comprehensive reforms. Countries with more fragmented institutional architectures or stronger reform-blocking capacities have tended toward incremental parametric adjustment. In all cases, however, the direction of travel has been the same: increasing effective retirement ages, moderating benefit growth, and strengthening the actuarial foundations of public pension systems.

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